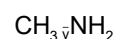
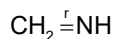
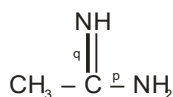
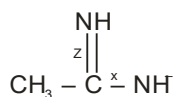


CHEMISTRY

TARGET : JEE- Advanced

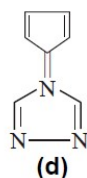
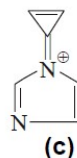
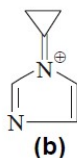
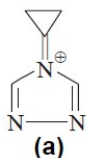
CAPS-6**GENERAL ORGANIC
CHEMISTRY****SINGLE CHOICE QUESTIONS**

1. Correct order of bond length of p, q, r, x & y given in following compound is



- (A) $y > x > p > q = z > r$ (B) $y > p > x = z > q > r$
 (C) $p > y > x = z > q > r$ (D) $p > y > x > q = z > r$

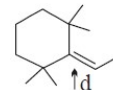
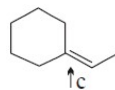
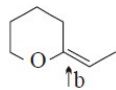
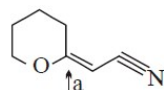
- 2.



Correct order of basic strength of this compound is

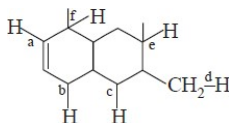
- (A) $d > b > c > a$ (B) $c > b > a > d$ (C) $c > a > b > d$ (D) $a > d > c > b$

3. Compare rotational energy barrier for the indicated bond.



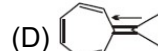
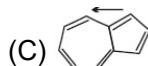
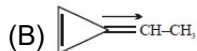
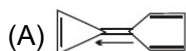
- (A) $d > c > b > a$ (B) $a > b > c > d$ (C) $c > d > a > b$ (D) $b > a > d > c$

4. Compare C-H bond energy.

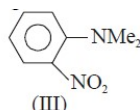
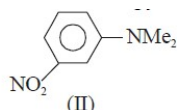
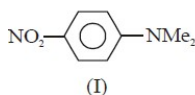


- (A) $a > b > c > d > e > f$ (B) $a > d > c > e > b > f$
 (C) $f > e > d > c > b > a$ (D) $f > b > e > c > d > a$

5. Which of the following is correct regarding direction of dipole moment?



6. The correct order at resonance energy in following.



- (A) $\text{II} > \text{I} > \text{III}$ (B) $\text{III} > \text{II} > \text{I}$ (C) $\text{I} > \text{II} > \text{III}$ (D) $\text{II} > \text{III} > \text{I}$



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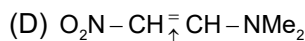
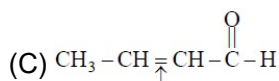
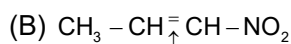
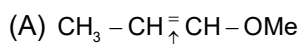
OFFICE ADDRESS : PLOT NUMBER 35, GOPALPURA BYPASS ROAD, NEAR RIDDHI SIDDHI CIRCLE,
10-B SCHEME, TRIVENI NAGAR, GOPAL PURA MODE, JAIPUR, RAJASTHAN 302020

Mob. 8888000021, 7410900901, 7410900906, 7410900907, 7410900908

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Page # 36

7. Which molecule has least rotational energy barrier for bond indicated by $\uparrow$.



8. Statement-1 :  is more stable carbocation than .

Statement-2 : +I effect of $\text{Me}_2\text{CH}-$ is more than that of $\text{Me}-$

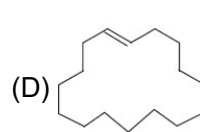
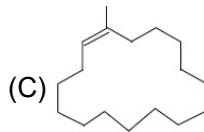
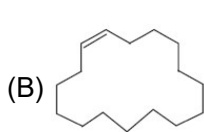
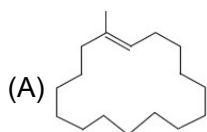
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

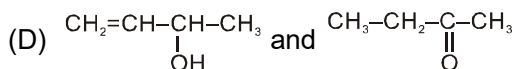
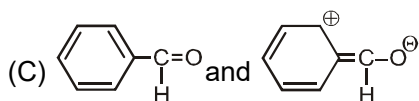
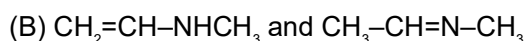
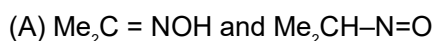
(D) Statement-1 is false, statement-2 is true.

9. Which of the alkene will have maximum Heat of hydrogenation

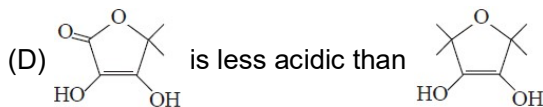
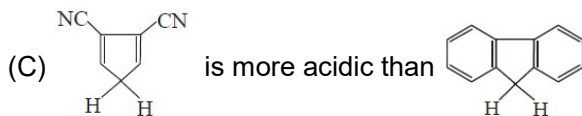
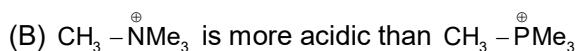
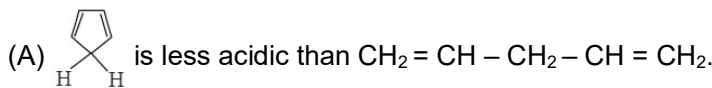


MULTIPLE CHOICE QUESTIONS

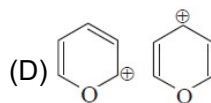
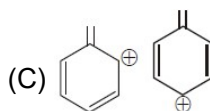
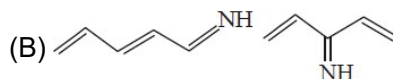
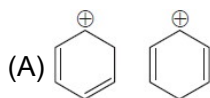
10. The tautomeric pairs are



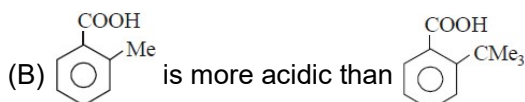
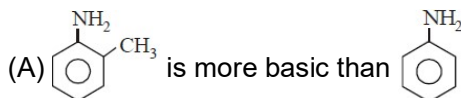
11. Which of the given statement is/are not correct?



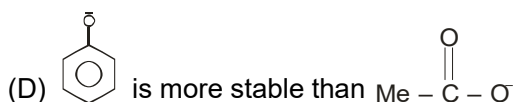
12. Which of the following pairs has 1st having more resonance energy than 2nd.



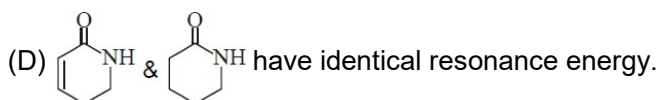
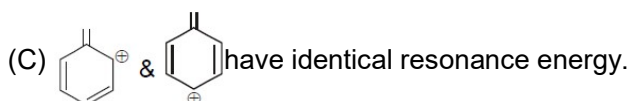
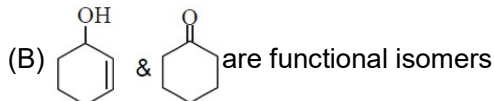
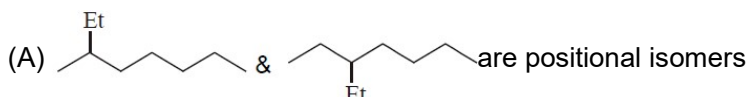
13. Select incorrect statement from the following.



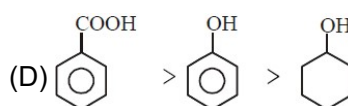
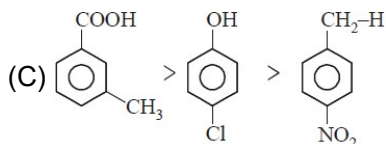
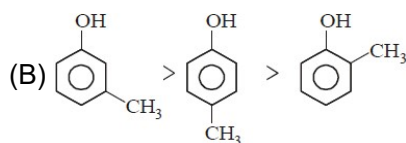
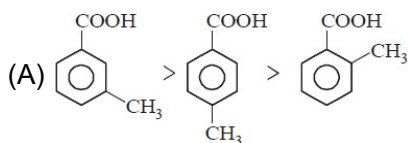
(C) $\text{HC} \equiv \text{CH}$ is more acidic than NH_3



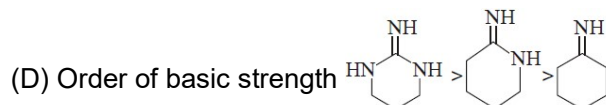
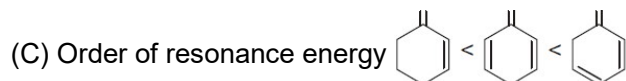
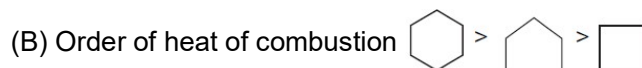
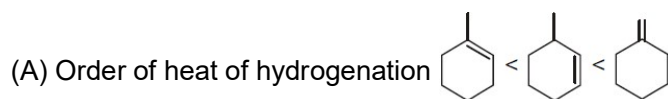
14. Select the correct statement.



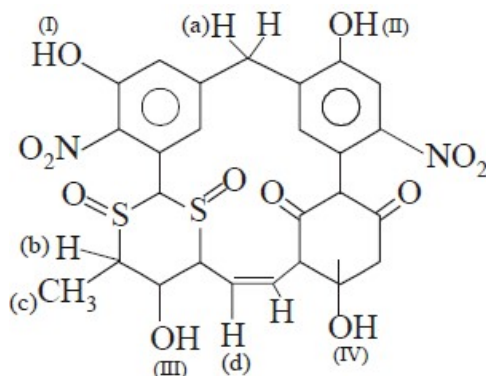
15. Which of the following is / are correct order for acidic strength.



16. Select the correct order among the following:



Paragraph for question nos. 17 to 19



17. The correct order of acidic strength of underlined H is

- (A) III > I > II > IV (B) I > II > IV > III (C) II > I > III > IV (D) I > II > III > IV

18. Total number of chiral centre present in the molecule is

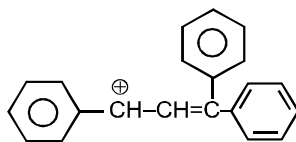
- (A) 6 (B) 7 (C) 9 (D) 10

19. The correct order of bond energy of the marked C – H bond homolysis is:

- (A) a > b > c > d (B) d > c > b > a (C) c > b > d > a (D) d > c > a > b

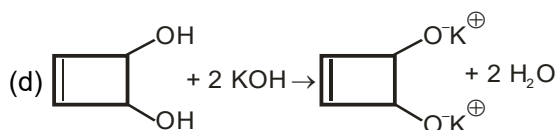
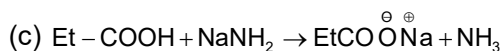
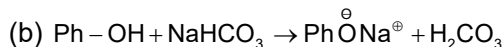
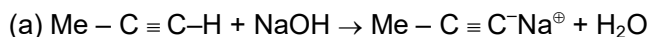
INTEGER ANSWER TYPE

20. Find the total number of the position where positive charge can be delocalized by true resonance

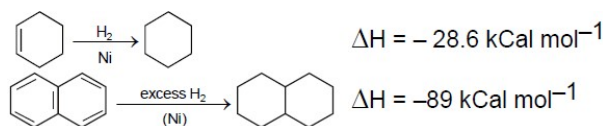


(Excluding the given position)

21. Fill 1 in OMR sheet if the reaction is favoured in forward direction and fill 2 in OMR sheet if the reaction is favoured in backward direction for example if all the reactions are favoured in forward direction fill 1111 as answer, and if only first two are favoured in forward direction fill 1122 in OMR sheet.



22. (a) Use the following data to answer the question below:



naphthalene

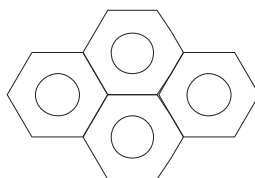
Calculate the resonance energy of naphthalene in kCal/mol.

- (b) How many grams of CO_2 is released when 84 gm of sodium bicarbonate is treated with excess of acetic acid.

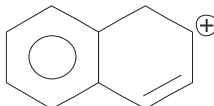
If answer of part (a) is xy and part (b) is pq then present your answer as xypq in the OMR sheet (round of your answer to nearest integers).

For example: if Answer of (a) is 12 and (b) is 13 you will fill 1213 in OMR sheet.

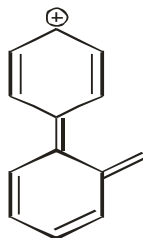
23. How many neutral (Benzoid) resonating structures are possible for.



24. Find out total no. of monoionic resonating structures of the following (Including given)



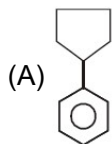
25. Find out total no. of monoionic resonating structures of the following (Including given)



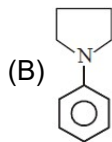
26. Match the Column.

Column I

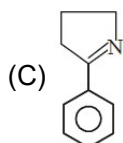
Column II



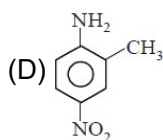
(P) Group attached to the benzene ring is +M effect showing group



(Q) Group attached to the benzene ring is -M effect showing group



(R) Group attached to the benzene ring is +H showing group.



(S) Group attached to the benzene ring is -I effect showing group

(T) Compound is aromatic

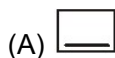
27.

Column I

Column II

(Compounds)

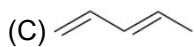
(Magnitude of heat of hydrogenation)



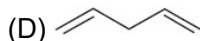
(P) 52.5 Kcal / mole



(Q) 45.9 Kcal / mole



(R) 26.6 Kcal / mole



(S) 30.7 Kcal / mole

Answer Key CAPS-6

1. (B)	2. (B)	3. (A)	4. (B)	5. (B)
6. (C)	7. (D)	8. (D)	9. (B)	10. (AB)
11. (ABD)	12. (BD)	13. (ABD)	14. (BC)	15. (BCD)
16. (BCD)	17. (D)	18. (C)	19. (B)	20. (10)
21. (2212)	22. (5444)	23. (6)	24. (7)	25. (13)
26. (A) R, T (B) P, S, T (C) Q, S, T (D) P, Q, R, S, T				
27. (A) S (B) R (C) Q (D) P				